



Faculty of Computing
SECP2523 – DATABASE
SEM 1 - 2024/2025

INITIAL SYSTEM PROPOSAL

KADA Cooperative System

Group Name: Data Alchemist

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Stakeholder: Lembaga Kemajuan Pertanian Kemubu
(KADA)

Representative Name	Date Interviewed
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PROPOSED SYSTEM BACKGROUND

1. Introduction

Currently, all of KADA's procedures are manual, based on the interview section, we found out that the applicants must complete the form and submit it in person. This manual method takes a lot of time and is again easy to make mistakes. After that, The clerks need to manually enter data and handle applications, this problem resulting in inefficiencies and delays.

The approach that we have come out with is based on the problem of the system. We wanted to digitize these processes by creating a computerized system that allows users to complete their registration and loan applications online. The new system will feature a user-friendly interface where candidates will be able to electronically fill out forms, upload documents, and enter their personal information. Authorized KADA staff will have efficient access to this securely stored data in a centralized database for review and processing. The efficient review process, automated notifications, and application status tracking features of the loan application system would greatly shorten the time needed for approvals and increase applicant satisfaction.

The implementation of this online system will bring numerous benefits to KADA and its members. It will minimize human data input errors, cut down on paperwork, and improve operational efficiency. With faster processing times and more flexibility, the new system will increase member satisfaction and perhaps get more people to apply for loans and join the cooperative. This transformation will not only simplify administrative processes but also contribute to the growth and success of KADA.

The difference between our system and the existing system which is KOPETRO is that our goal is to ensure that all the users can easily use our website even though they don't have any knowledge about IT, especially for elderly. Our interface will be simplified and straightforward so that they can easily understand how to navigate it. Other than that, we will also ensure that our website can be used everywhere by having the lightweight design and fast loading page since some of the user live in area with limited internet connectivity

2. The Existing System

KADA (Lembaga Kemajuan Pertanian Kemubu) is an agriculture organization in Kelantan. The workers under KADA are being given the opportunity to become cooperative members of KADA and make a loan under KADA too. But, the current system, or the as-is system in KADA to provide such services is being done manually and there is where we saw that it has very low efficiency and productivity. According to the list below, are the services that KADA provided to their staff currently.

1. Registration to become a membership of cooperative of KADA

In the current system, all the registration procedures have to undergo a manual process, such as filling in the paper form. For example, users must pick up the form in person at the headquarters in order to fill it in. In the registration process, most of the personal information of the users that are private and sensitive will be collected. Then, the users have to submit it manually also by submitting it at the headquarters. After the filled in and submitting process is done, then the registration paper would be further reviewed by the staff itself manually whether the users are qualified to become their member or not.

2. Applying a loan in cooperative of the KADA

Same as the registration process to become a member of KADA, applying for a loan is also by filling in the paper form manually. The difference is in applying for a loan form, it would ask more detail about the user income information and guarantor. After the users have submitted it into the headquarters, it would be reviewed by the staff and the approval would be given by the higher position. The approval is given by conducting a monthly meeting between the staff and the higher position.

The current system has a lot of problem such as :

1. Loss of information

Most of the records are stored in paper such as the registration documents and also the loan documents. Paper documents can be easily damaged by water, fire or mishandling. Once damaged, the information on them may become unreadable or entirely lost. Other than that, paper documents can be misplaced and lost due to human error.

1. Accumulation of data when access or update

The clerk must locate the paper back in order to update the data, which is a difficult operation because they must scan and identify each paper in their safe in order to locate the particular one. As a result, it will take longer to finish a single task—updating the user's information

1. Highly dependent on human

Using paper forms greatly reduces productivity and efficiency because handling, filing, and finding these documents take a lot of time and effort. The dependence on paper slows down operations, causes delays, and keeps employees from focusing on more important tasks, ultimately making the organization less productive.

1. Difficulty in distributing and receiving the application

In order to obtain the form and submit it, applicants must make an extensive journey to the headquarters. They are certain to be burdened by this since they will need to find time to travel to the headquarters and spend money on transportation.

The proposed online system should replace the existing manual system because it effectively addresses critical issues such as data loss, inefficiency, human dependency, and limited accessibility. Currently, paper-based records are vulnerable to damage, loss, and misplacement, while the new system securely stores data electronically, ensuring better protection through backups and encryption. Manual processes require clerks to spend excessive time locating and updating records, whereas the digital system allows instant access and streamlined management, significantly improving productivity. Additionally, the reliance on human labor in the manual system slows operations, but automation in the proposed system reduces workload and accelerates application processing. For applicants, the current system requires physical visits to the headquarters, which is inconvenient and costly, whereas the online system allows users to submit forms remotely, anytime, with immediate confirmation, saving time and travel expenses. Members of the Board of Directors (ALK) also benefit from remote access to applications, enabling faster reviews and approvals. Overall, the proposed system enhances efficiency, accuracy, and user satisfaction, making it a superior solution that should replace the outdated manual system.

Table 1: Comparison of existing systems

<i>Features</i>	<i>Existing system (manual)</i>	<i>Proposed system (online)</i>
<i>Data Storage</i>	<i>Paper-based records prone to damage, loss, or misplacement.</i>	<i>Centralized, secure digital storage with backups and encryption.</i>
<i>Data Accessibility</i>	<i>Manual search through physical files, time-consuming and inefficient</i>	<i>Instant access to data through a searchable, user-friendly interface.</i>
<i>Application submission</i>	<i>Requires in-person visits to headquarters for form submission</i>	<i>Online form submission accessible from anywhere with an internet connection</i>

3. The Proposed System

The system will serve as a cutting-edge substitute for the outdated paperwork system. Thus, the system will perform the same basic tasks as the original paperwork system, including handling the registration form for the new member and also the registration form to apply for a loan for the member of the organization. On the other hand, The shift from physical documentation to an online form is the main upgrade for the system. In the past, the organization primarily employed paper in their operations to store user information and to verify users during the loan application and registration processes. Initially, the current system only had three primary user views: the applicator, Ahli Lembaga Pengarah (ALK), and the clerk. However, after a few discussions, we concluded that the admin roles would be essential for the new online system.

Firstly, from the view of the applicants, The new system will definitely be beneficial for the user as the online system offers many benefits for the applicants. With the current paper system, people have to travel to the headquarters to get and submit forms, which takes a lot of time and money. This can especially be difficult for those who live far away or have busy schedules. In contrast, the online system allows people to access and submit forms anytime and from anywhere with an internet connection. This means they don't need to make a special trip, saving both time and travel costs. The online system is available 24/7, making it much more convenient. It also provides instant confirmation when an application is submitted, so applicants don't have to worry whether their forms are received by KADA or not. Additionally, online forms often help prevent mistakes by checking for missing or incorrect information before submission. Overall, the online application system makes the process faster, easier, and cheaper for everyone involved.

Secondly, for the members of the Board of Directors (ALK), their job of checking membership and loan applications becomes much easier with the new online system. Before, this process was slow and difficult because they had to be at the office and deal with lots of paperwork, which limited their flexibility and caused delayed decisions. With the online system, ALK members can do their work from anywhere and at any time, without needing to travel or handle physical documents. This means they can approve applications faster and more easily. The system also provides instant updates and notifications, making their work more efficient. As a result, the organization becomes much more productive in completing tasks that used to take days or weeks in a much shorter time. This makes the work of ALK members easier and improves the overall efficiency and speed of the organization.

The discussion revolves around creating a user-friendly system interface for clerks to handle applications smoothly. This interface allows clerks to easily access applicant details and payment records, and to update application statuses promptly. A key task for clerks is to review submitted forms, ensuring they meet the requirements set by KADA. With the system's filtering features, clerks can quickly sort through applications, accepting those that meet criteria and rejecting incomplete or non-compliant ones. By diligently updating database records, clerks maintain accurate information for organizational use. This streamlined process helps clerks efficiently manage applications, contributing to the organization's overall effectiveness.

Lastly, for the administrative, their role in this new proposed system is to maintain the data and also the system itself. The administration needs to ensure that the system can run smoothly without any problem. Other than that, they also need to ensure that every application information is recorded in the database so that the data is safely secure.

4. Glossary

- 1)** KADA (Lembaga Kemajuan Pertanian Kemubu): An agriculture organization based in Kelantan that provides loans and cooperative membership opportunities for its staff.
- 2)** ALK (Ahli Lembaga Pengarah): Board of Directors responsible for reviewing and approving membership and loan applications.
- 3)** Centralized Database: A digital storage system that securely keeps all data in one location, accessible to authorized users.
- 4)** Encryption: Securing data by converting it into a code to prevent unauthorized access.
- 5)** User Interface (UI): The visual part of the system through which users interact with the application.
- 6)** Automation: Using technology to perform tasks with minimal human intervention, improving efficiency and accuracy.
- 7)** Filtering Features: Tools that allow clerks to sort and manage applications based on specific criteria.
- 8)** Paper-based System: Traditional manual processes involving physical documents for data storage and operations.
- 9)** AS-IS system: Current or existing system in its present state, including its processes, features, and limitations.

5. Bibliography

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